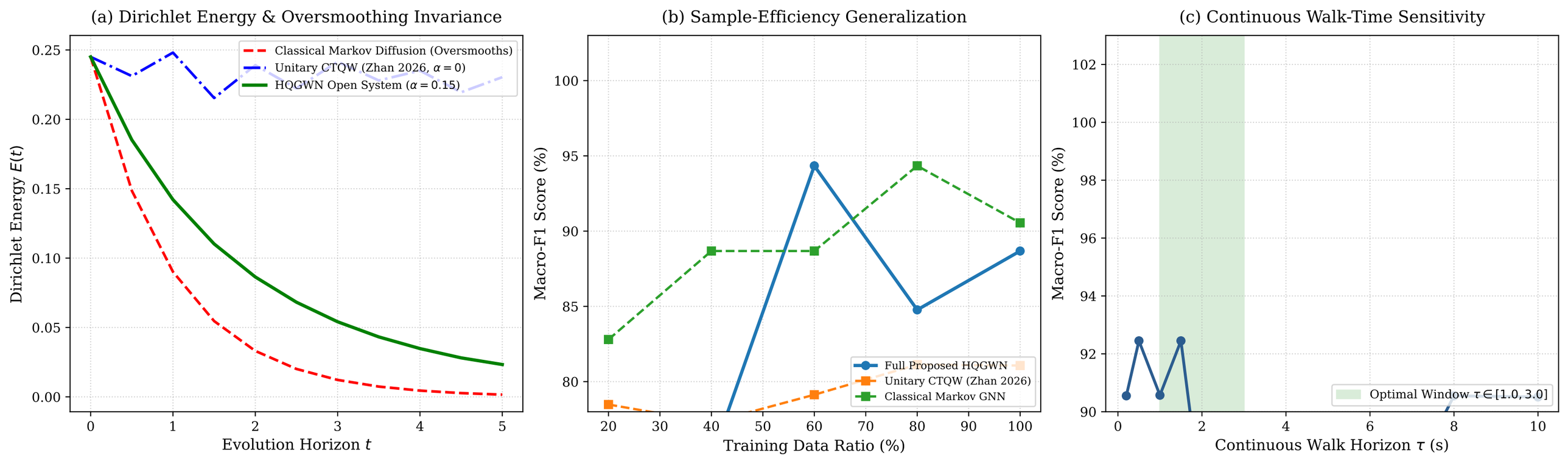
(a) Dirichlet Energy & Oversmoothing Invariance
Classical Markov Diffusion (Oversmooths)
Unitary CTQW (Zhan 2026, $\alpha = 0$)
HQGWN Open System ($\alpha = 0.15$)
Dirichlet Energy $E(t)$
Evolution Horizon t
0.25
0.20
0.15
0.10
0.05
0.00
0
1
2
3
4
5
(b) Sample-Efficiency Generalization
Full Proposed HQGWN
Unitary CTQW (Zhan 2026)
Classical Markov GNN
Macro-F1 Score (%)
Training Data Ratio (%)
100
95
90
85
80
20
30
40
50
60
70
80
90
100
(c) Continuous Walk-Time Sensitivity
Optimal Window $\tau \in [1.0, 3.0]$
Macro-F1 Score (%)
Continuous Walk Horizon τ (s)
102
100
98
96
94
92
90
0
2
4
6
8
10